

B. Tech. First Semester (2024-25)

Tutorial Sheet – II

Course: Physics – I (15B11PH111)

Assignment 2: Derivation of wedge-shaped films

Section A: Mandatory Questions

- 1.a) What is the value of phase change upon reflection from medium 1 (refractive index μ_1) to medium 2 (refractive index μ_2) if (i) $\mu_1 > \mu_2$ and (ii) $\mu_1 < \mu_2$?
b) What are the fringes of equal width and fringes of equal inclination?
c) Write the expression for fringe width in interference from thin films. [CO 1]
- 2.a) Using Stoke's relations, show that the excessively thin film appears perfectly dark, when seen by reflected light.
b) Why is the central fringe of Newton's ring formed by reflected light dark?
c) Why are complementary colours visible in interference patterns in thin film for reflected light and transmitted light? [CO 2]
3. Light of wavelength $6000 \text{ \AA}$ falls normally on a thin wedge-shaped film of refractive index 1.4, forming fringes that are 2 mm apart. Find the angle of the wedge. [CO 2]
4. In the Newton's rings arrangement, the radius of curvature of the curved surface is 50 cm. The radii of the 9th and 16th dark rings are 0.18 and 0.2235 cm, respectively. Calculate the wavelength. [CO 2]
5. A soap film of refractive index $\mu = 1.34$ is surrounded by air. Under normal illumination, if a region of the film strongly reflects the wavelength $\lambda = 804 \text{ nm}$, what is the minimum thickness of the film? [CO 3]
6. An equiconvex lens is placed on another equiconvex lens. The radii of curvature of the two surfaces of the upper lens are 50 cm and those of the lower lens are 100 cm. The light waves reflected from the upper and lower surface of the air film (formed between the two lenses) interfere to produce a fringe pattern. If the monochromatic light of wavelength $6000 \text{ \AA}$ is used in the experiment, calculate the radii of the 4th and 16th dark rings. [CO 3]
7. Suppose that the light source in the Newton's ring arrangement emits two wavelengths λ_1 and λ_2 , such that the m -th dark ring due to λ_1 coincides with the $(m + 1)$ -th dark ring due to λ_2 . If R is the radius of curvature of the lens, show that the radius ρ_m of the m -th dark ring due to λ_1 is given by

$$\rho_m = \sqrt{\frac{\lambda_1 \lambda_2 R}{\lambda_1 - \lambda_2}}. \quad [\text{CO 4}]$$

8. Consider the formation of Newton's ring with a bi-chromatic light source emitting two closely spaced wavelengths λ_1 and λ_2 . Show that if plano-convex lens is raised by a distance $t_0 = \frac{\lambda_1 \lambda_2}{4(\lambda_1 - \lambda_2)}$, the fringe pattern would disappear. If we continue to move the lens upward, show that the fringe pattern would reappear when lens is raised by a distance $2t_0$. Assume that the light source is placed normally above the lens. [CO 4]

Section B: Practice Questions

1. In the Newton's ring arrangement, the radius of curvature of the curved surface of the plano-convex lens is 100 cm. For $\lambda = 6 \times 10^{-5}$ cm, what will be the radius of the 9th bright ring? [CO 2]
2. White light is reflected normally from a soap film of uniform thickness in air. An interference maximum is observed at $\lambda_1 = 0.6 \mu\text{m}$ and a minimum at $\lambda_2 = 0.45 \mu\text{m}$ with no minimum or maximum in between. Assuming $\mu = 1.33$ for the film, calculate the thickness of the film. [CO 3]
3. A microscope of refractive index $\mu = 1.55$ is to be coated with a MgF_2 film ($\mu = 1.38$) to increase transmission of normally incident light $\lambda = 550$ nm. What should be the minimum thickness of the film? [CO 4]
4. In the Newton's rings arrangement, if the incident light consists of two wavelengths of 4000 and 4002 Å, calculate the distance (from the point of contact) at which the rings will disappear. Assume that the radius of curvature of the lens is 400 cm. [CO 4]